Adapting Open-Source Artificial Intelligence to Combat Casualty Care: The Case Study of our Joint Human and Skin Segmentation Neural Network

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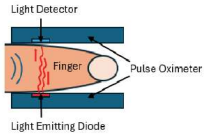


Objectives

- Create accurate skin segmentation masks from video at various ranges, positions, and lighting for warfighters in austere military conditions
- Use a single encoding network for accurate identification of both skin and body and key point detection for computational efficiency
- Improve non-contact vitals measurement from video via robust skin segmentation

Background

- Skin segmentation is an incredibly useful tool for image-based Photoplethysmography extraction, which is used for heart-rate measurements
- Identifying skin is often approached via color thresholding or with facial key points, both of which fail in camouflage or obscuration
- Deep-learned convolutional models are promising, however most open-source models are not suitable for a wide range of poses or environments and fail in military conditions
- Custom deep learning models for skin segmentation have shown promise when paired with human feature detection



Typical Method for PPG Extraction

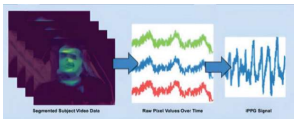
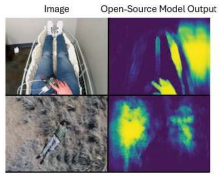


Image-Based PPG Extraction via Skin Mask



Open-Source Skin Segmentations: *Skin Segmentation Models often fail when applied outside the data they were trained on, such as with targets in multiple ranges and in austere military conditions*



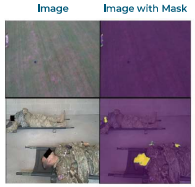
Drone Mounted Collection System: *Single camera mounted on drone, designed for covering large areas and identifying patients in difficult terrain*

Methods

- Open-Source Data bolstered with privately sourced military relevant data
- The mixed training data included a wide range of different environments as the privately collected data supplied images of subjects outside and various positions and lighting not often found in datasets
- Augmentation included color, rotation, and insertion augmentations including adding red shapes to the data to reduce false alarms
- Ground truth was created iteratively with a combination of color thresholding, convolutional segmentation via a U-Net encoder, bootstrapping, and finally hand sorting of accurate masks



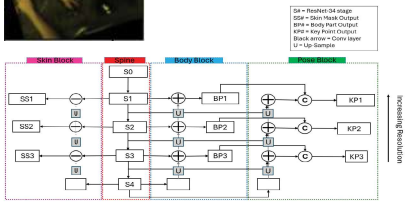
Open-Source Image Collection with Ground Truth Masks



Privately Collected Data at a Range of Distances



Privately Collected Image with Red Shape Insert for False Alarm Reduction during Training of Network



Branching Combined Skin and Human Detection Architecture

- Skin Segmentation performance was increasing by appending the skin weights onto an encoding spine trained for human detection
- The Resnet-34 spine branches off to support body part segmentation and then subsequently key point detection
- Weights for skin segmentation were appended onto the spine after the body and key point weights were separately trained
- The architecture is relatively lightweight and allows for one, two, or all three of the different blocks to run in a single forward pass

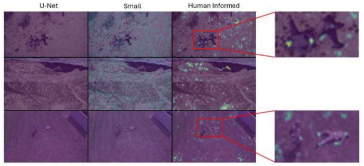
Results

Timing and performance results compare three different networks:

- **Small Model:** 4 convolution layers, very small and fast, features are mainly color based
- **U-Net Model:** Used for ground truth, commonly found in literature, deep-architecture
- **Human-Informed Model:** Skin layers appended to larger human-focused model

	Small Skin Model	U-Net Skin Model	Human Informed: Just Skin Branch	Human Model w/o Skin Branch	Small + Human Model	U-Net + Human Model	Human Informed: All Branches
Median Forward Pass (sec)	0.0021	0.0044	0.0077	0.0088	0.011	0.013	**0.0097**
Parameters (MB)	0.32	13.12	-	88.01	**88.03**	101.13	91.22

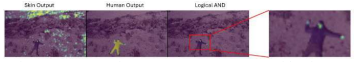
Table Comparing Speed and Memory for Different Models



Qualitative Analysis of Model Performance on Drone Imagery Taken in Military Conditions

- The Small Model while the fastest and smallest in isolation, is slower when paired with the need for a human detection network
- The Human-Informed model has the fastest combined forward pass for both human and skin detection

- U-Net has very few false alarms, but struggles to segment at various ranges
- Small has many low-level false alarms, and often fails to segment limb
- Human-Informed segments all limbs effectively, but struggles with false alarms
- False alarms are mitigated with the human detection



Using Human Detection Outputs For False Alarm Mitigation

	L1 Error	IoU	Precision	Recall	F1 Score	Forward Time with Human Detection (s)	Pass with Human Detection	Total Parameters with Human Detection (MB)
Simulated Disaster Victim Dataset (SDV1) Mean Performance Metrics								
U-Net Model	**0.019**	0.42	0.57	**0.8**	0.66	0.013	101.13	
Small Model	0.12	0.18	**0.76**	0.18	0.28	0.011	**88.03**	
Human Informed Model	0.021	**0.43**	0.63	0.7	**0.66**	**0.0097**	91.22	

Table Showing Quantitative Performance of Skin Segmentation on a Hold Out Set with Simulated Disaster Victims

- Human-Informed model performs has the best IoU and F1 score on hold-out images with hand-truthed masks
- Small model fails in unideal green lighting conditions and over segments on regions with hair
- Human-Informed model closely measures Heart Rate in all three lighting conditions



Comparing Skin Segmentations between Small and Human Informed Models with Subjects in Unideal Lighting Situations



Human Informed Skin Segmentation Model Outputs on a Hold Out Set with Simulated Disaster Victims

	Small Model	Human-Informed Model	Ground Truth Measured by ECG
Mean Red Light HR (BPM)	79.4+-2.7	70.1 +-2.4	77.9 +- 1.9
Mean Green Light HR (BPM)	48.3+-15.7	76.6+-3.4	71.0 +-1.7
Mean Flashing Light and Movement HR (BPM)	74.9+-1.5	82.6+-3.4	71.0+-1.7

Table Comparing Non-Contact Heart Rate Monitor Accuracy on Subjects in Unideal Lighting Conditions

Conclusion

AI can change how armed forces approach telehealth and remote monitoring in austere conditions. With lightweight convolutional models, accurate autonomous triaging is possible. Joint human and skin segmentation enables identification of people and skin in diverse environments. These networks are essential for accurate vitals measurement such as heart rate monitoring. Several field tests were used to evaluate the performance of the human and skin network and performance was validated with images from vastly different contexts, angles, and poses in open-source datasets.

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